National Teachers College
629 J Nepomuceno, Quiapo, Manila, 1001 Metro Manila
Bachelor of Science in Information Technology
IT Elective 2

Final Project/Exam:
C# Application Development for Sustainable Development

Submitted by:

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BSIT 3.2

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I. Introduction

1.1 Project Overview:

The project, SILAKBO: A Women-Only Harassment Reporting and Support System, is a C# Windows Forms application designed to provide a safe and confidential platform for women to submit harassment reports and access support resources.

SILAKBO aims to empower women by giving them a secure way to report incidents, view the status of their cases, and access a directory of available support services. The system prioritizes user safety and ensures that all reports remain private and managed by administrators.

This project aligns with United Nations Sustainable Development Goal 5: Achieve gender equality and empower all women and girls, particularly Target 5.2 – Eliminate all forms of violence against women and girls in public and private spheres. By limiting access to women only, SILAKBO safeguards users' emotional well-being while providing fair and secure protection for vulnerable individuals.

1.2 Problem Statement:

Harassment and abuse continue to be serious issues affecting women and girls, yet many incidents go unreported due to fear of retaliation, social stigma, emotional trauma, or lack of trust in authorities. Existing reporting methods are often manual, inconsistent, or insecure, forcing victims to share sensitive information in environments where privacy cannot be guaranteed.

Additionally, women may feel hesitant to report incidents if men, including potential abusers can access the platform, further discouraging reporting and leaving abuse unaddressed.

There is a clear need for a secure, women-only digital platform where survivors can safely submit reports, monitor the status of their cases, and access verified support resources. SILAKBO fulfills this need by offering confidential reporting, centralized

support information, and a user-focused environment that empowers women and girls to take control of their safety and well-being.

II. Requirements & Analysis

2.1 Functional Requirements:

ID	Functional Requirements
FR1	**User Management:** The system performs CRUD operations on a relational database, including user registration, report submission, report viewing, and status updates.
FR2	**Confidential Reporting Module:** The system processes report submissions and manages case status updates through a dedicated service layer, ensuring proper handling of report data.
FR3	**Visualization:** The system displays a graphical chart summarizing report data based on case status (e.g., Pending, Ongoing, Completed).
FR4	**Security:** The system implements a hashed login mechanism to securely store and verify user credentials.

2.2 Non-Functional Requirements:

Category	Requirement
Performance	The system ensures efficient performance by using asynchronous operations (async/await) for loading reports and other data, preventing the user interface from freezing and maintaining smooth

	interaction during database operations.
Security	The system secures user data by implementing a hashed password mechanism during registration and login, ensuring that sensitive credentials are not stored in plain text.
Robustness	The system maintains reliability and stability by handling exceptions using try-catch blocks and recording errors in an external log file, allowing for proper debugging and error tracking.

2.3 Complexity Analysis:

Core Business Logic Methods

1. Submit Report (ReportService.Submit())
    - Inserts a new report into the database.
    - Time Complexity: O(1) – Insertion into a relational database is constant time per record.
    - Space Complexity: O(1) – Only the new report object is temporarily stored in memory.
2. Load Reports (ReportService.GetReportsByUser(userID))
    - Retrieves all reports for a user (or all reports for admin).
    - Time Complexity: O(n) – Scales linearly with the number of reports n in the database.
    - Space Complexity: O(n) – Stores all retrieved reports in a list in memory.
3. Update Case Status (ReportService.UpdateStatus(reportID, newStatus))
    - Updates the status field of a specific report.
    - Time Complexity: O(1) – Updating a single record by its primary key is constant time.
    - Space Complexity: O(1) – Only the modified status value is stored temporarily.

4. Backup Reports to JSON (ReportService.BackupReports())
    - Serializes all reports to a JSON file.
    - Time Complexity: O(n) – Each report must be read and written to JSON.
    - Space Complexity: O(n) – All report data is stored in memory during serialization.

III. Design Specification

3.1 N-Tier Architecture Diagram:

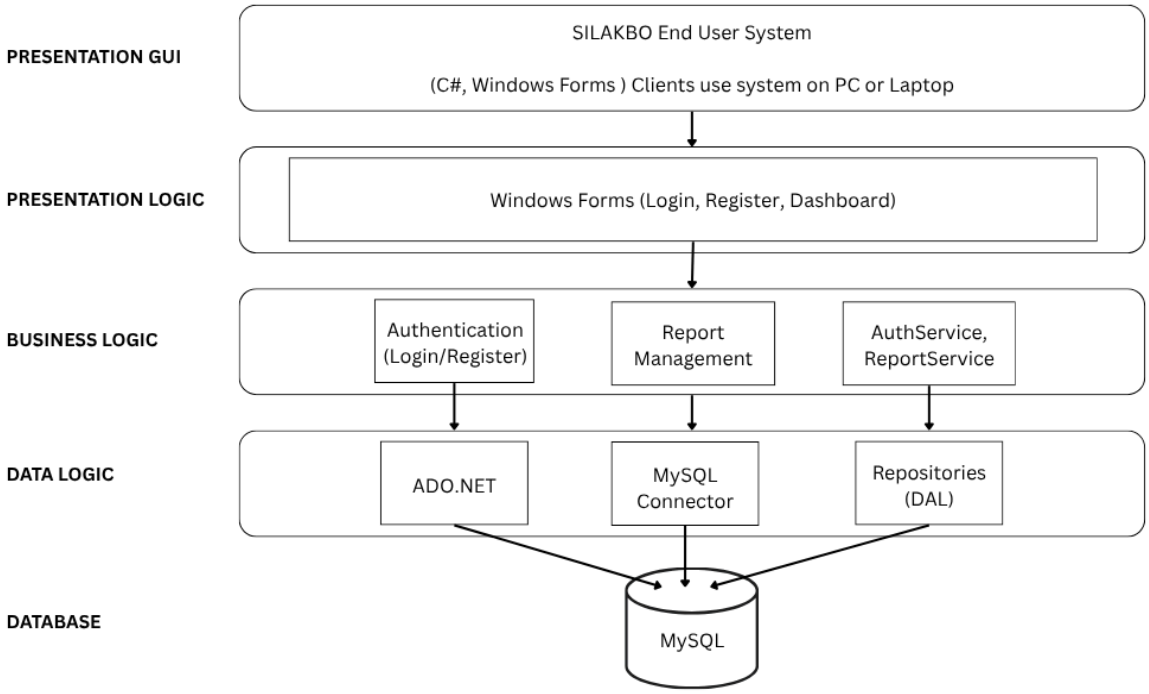


Figure 1: *N-Tier Architecture Diagram*

3.2 Entity Relationship Diagram (ERD):

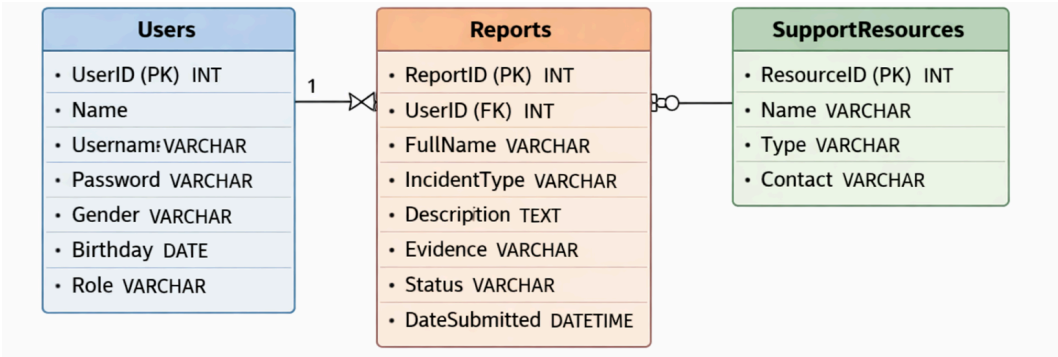


Figure 2: *Entity Relationship Diagram (ERD)*

3.3 Use Case & Data Flow Diagrams (DFD):

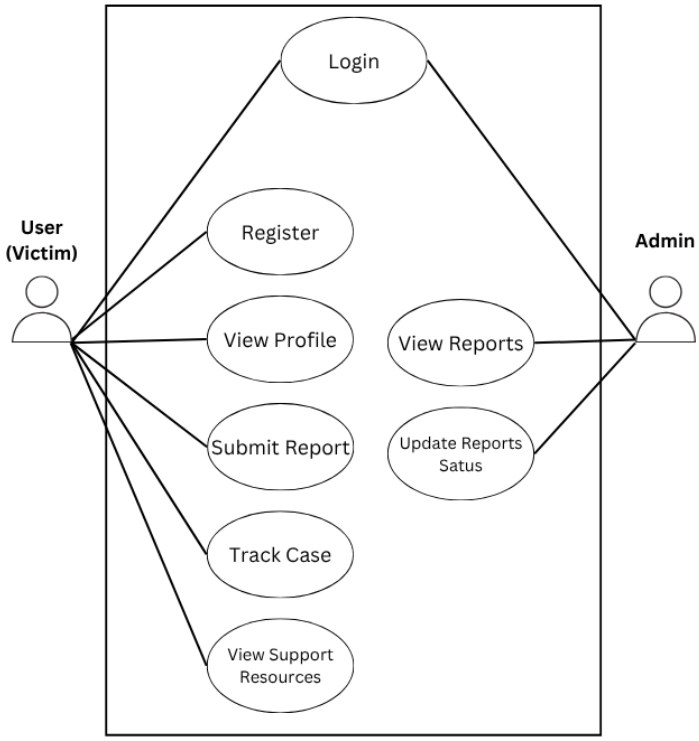


Figure 3:*Use Case Diagram*

Data Flow Diagram Level 0 (Context Diagram)

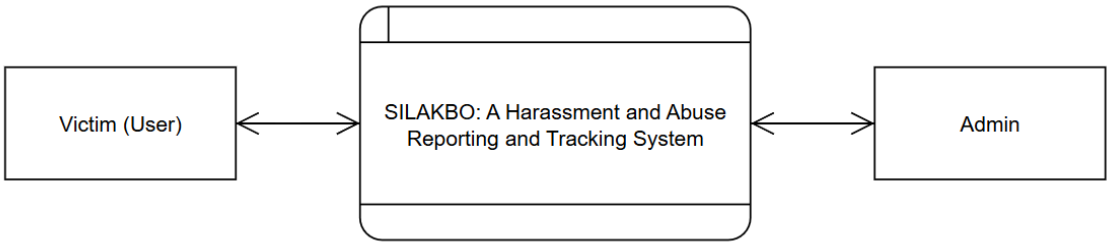


Figure 4: *Data flow diagram (Level 0)*

Data Flow Diagram Level 1

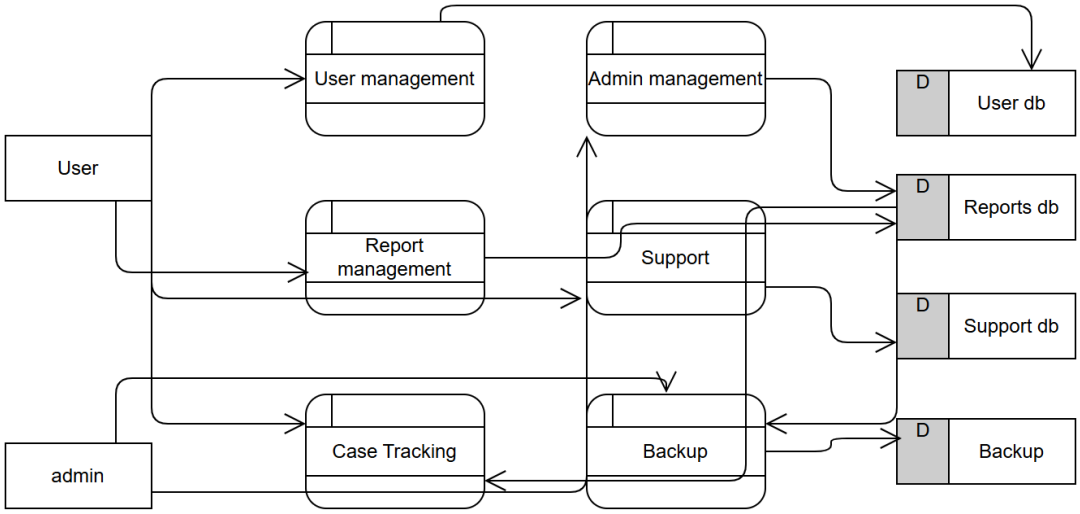


Figure 5: *Data flow diagram (Level 1)*

3.4 Algorithm Flowchart:

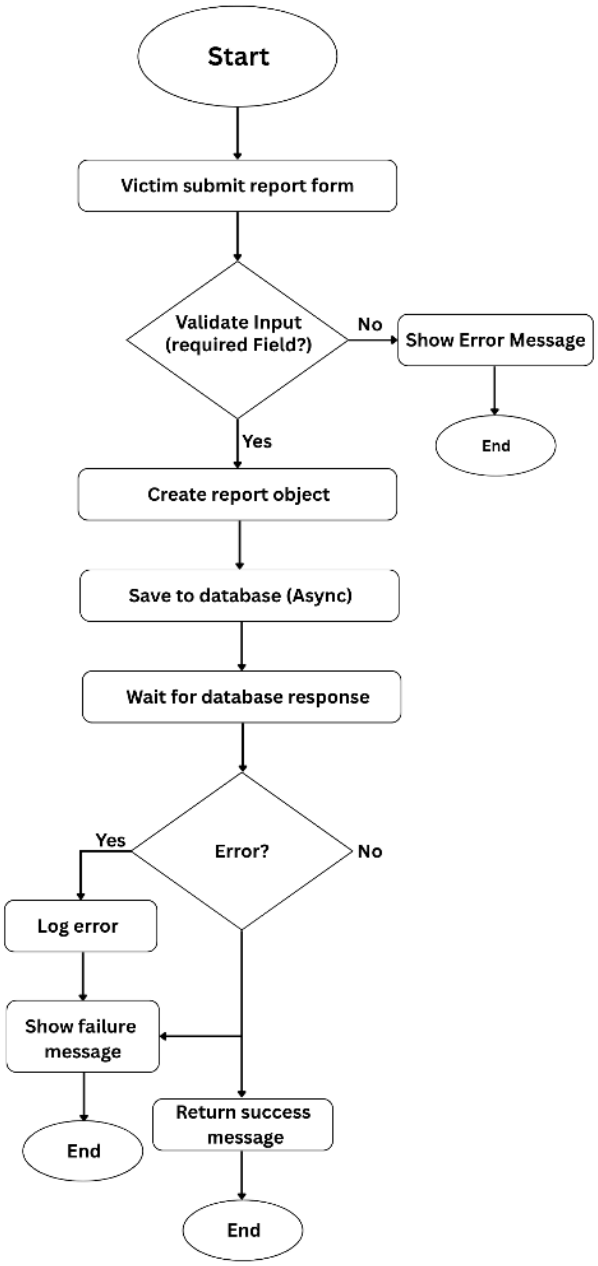


Figure 6: *Algorithm Flowchart*

IV. Individual Contributions

- A table detailing exactly which Layer or Module was implemented by which member.

Member	Assigned Layer / Module	Description of Work
Sarina Mae A. Laurel	Core Functionality / All Modules	Implemented main logic, integrated modules, handled report submission, case tracking, and analytics/reporting
Reiven Jan P. Sayno	Database & Admin Dashboard	Created and maintained database, optimized queries, and fixed errors in Admin Dashboard
Ahdlei Shane R. Villanueva	UI / Design	Designed the user interface, forms, layouts, and overall visual design

V. Appendices

- **Login Page -** Allows registered users to securely access the system using their credentials.

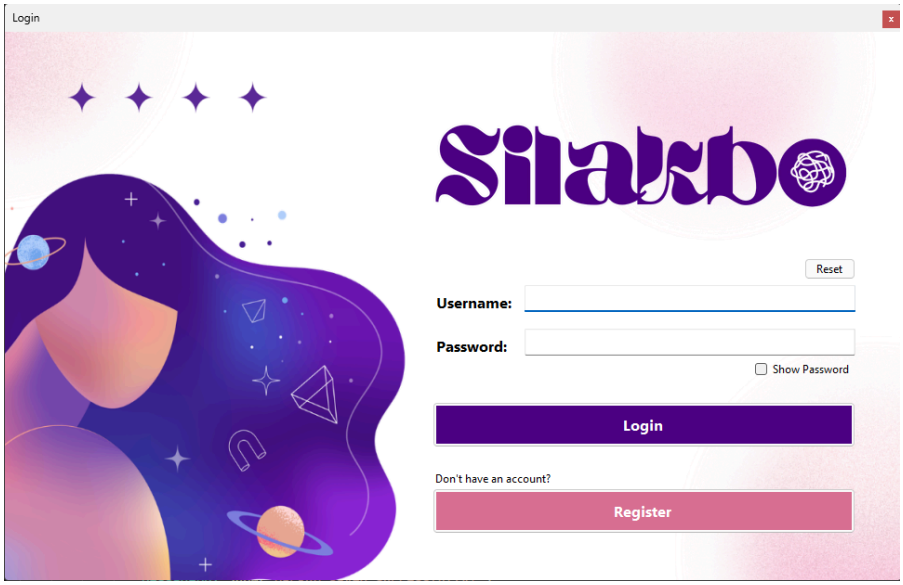


Figure 7: *Login Page*

- **Registration Page -** Enables new users to create an account by providing personal details and setting a password.

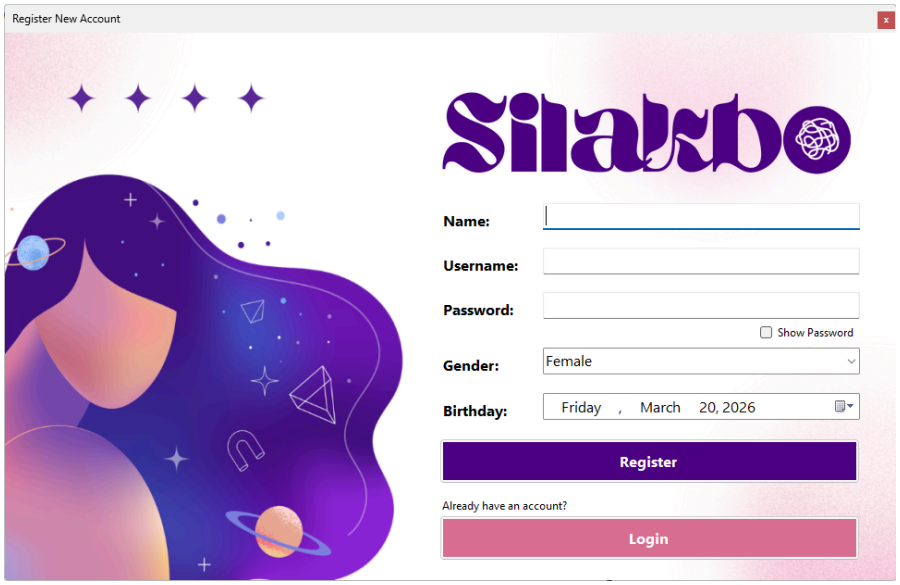


Figure 8: *Registration Page*

User/Victim Side

- **Victim Dashboard** - Serves as the main interface where users can access features such as profile viewing, report submission, case tracking, and support resources.

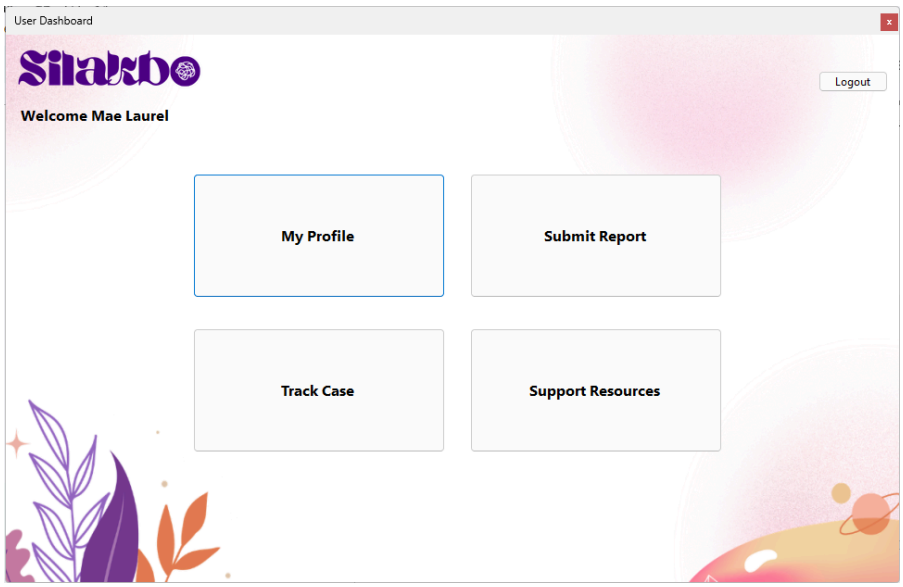


Figure 9: *User/Victim Dashboard*

- **Profile Page** - Displays the user's personal information and allows them to view their registered details.

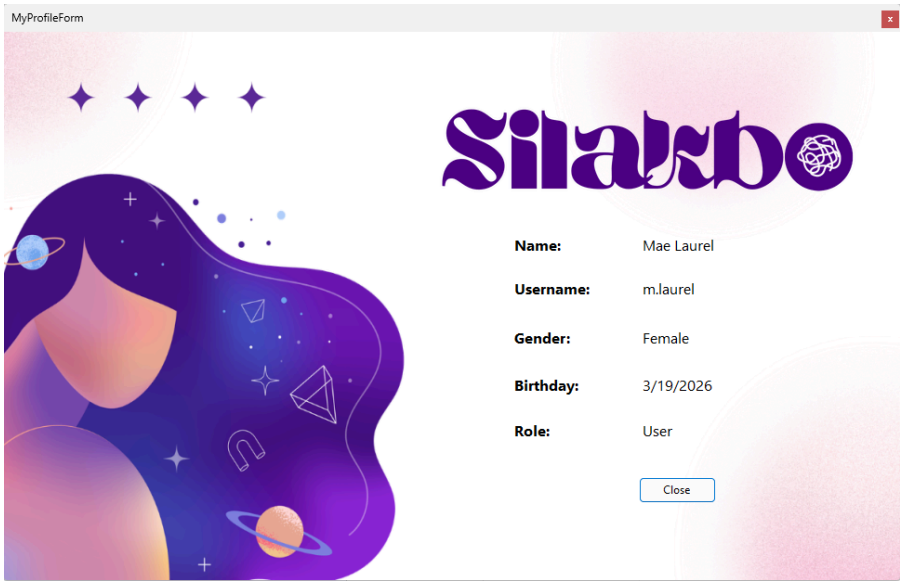


Figure 10: *Profile Page*

- **Harassment Report Page -** Allows users to submit detailed reports of harassment incidents, including descriptions and optional evidence files.

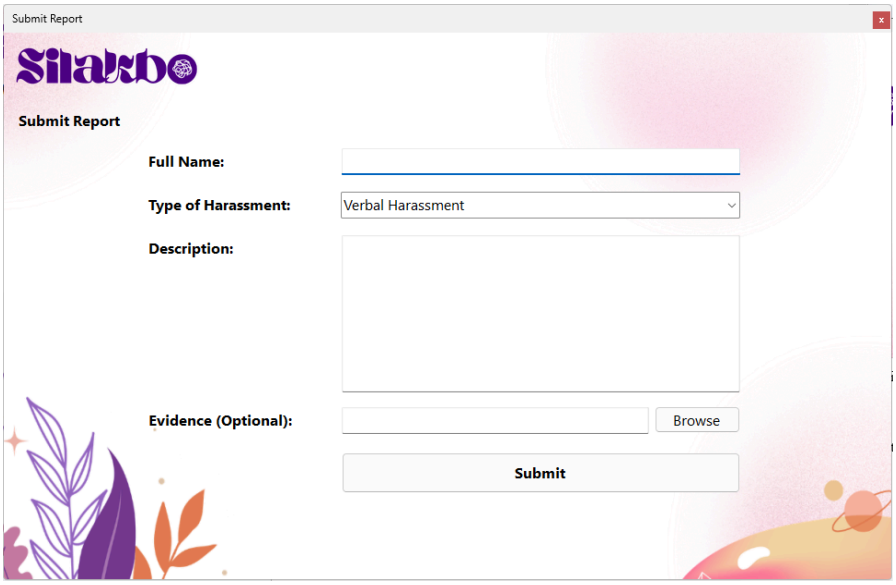


Figure 11: *Harassment Report Page*

- **Track Case Page -** Enables users to view and monitor the status of all reports they have submitted.

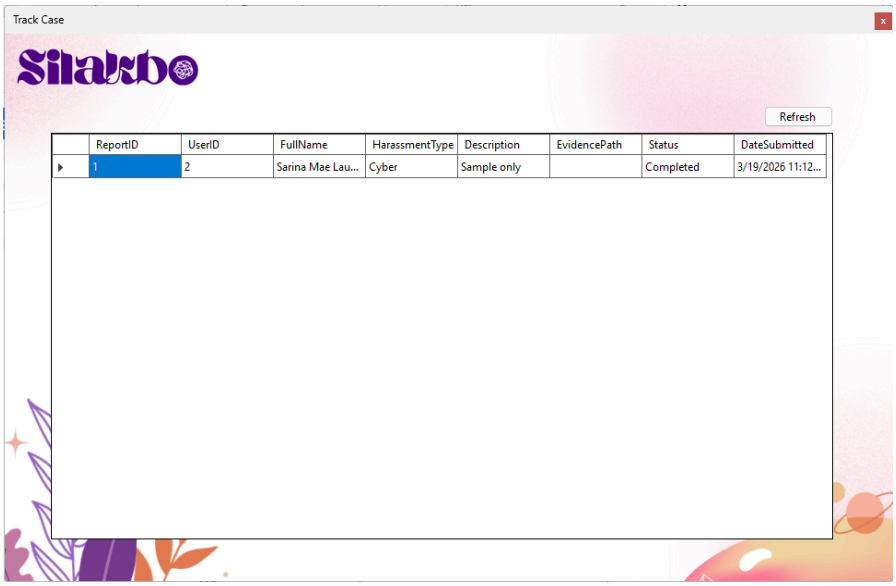
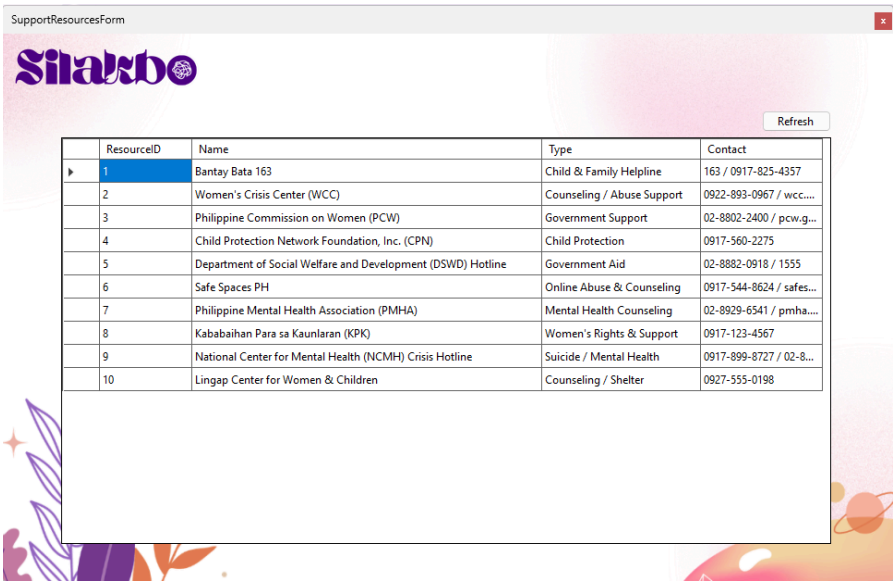


Figure 12: *Track Case Page*

- **Support Resources Page -** Provides a list of available support services with contact details to assist users in seeking help.

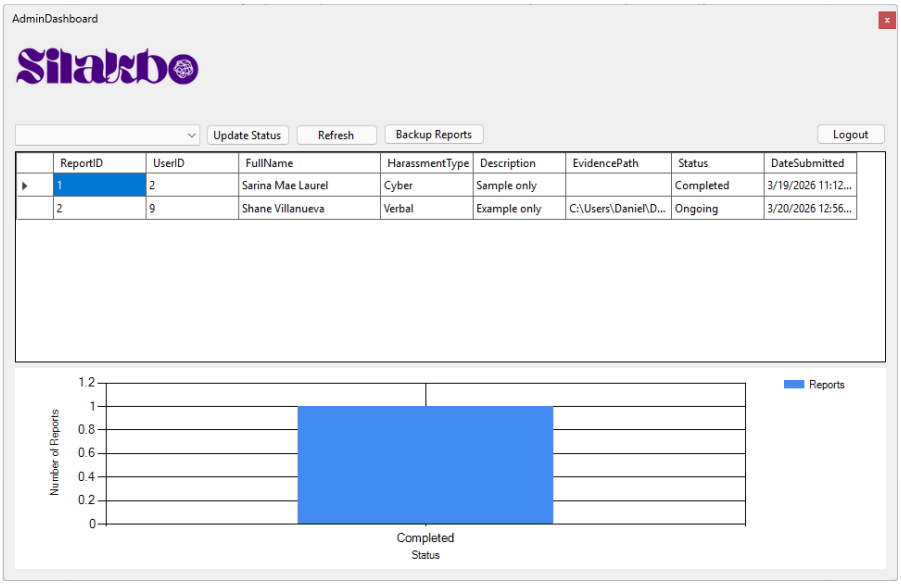


ResourceID	Name	Type	Contact
1	Bantay Bata 163	Child & Family Helpline	163 / 0917-825-4357
2	Women's Crisis Center (WCC)	Counseling / Abuse Support	0922-893-0967 / wcc....
3	Philippine Commission on Women (PCW)	Government Support	02-8802-2400 / pcw.g...
4	Child Protection Network Foundation, Inc. (CPN)	Child Protection	0917-560-2275
5	Department of Social Welfare and Development (DSWD) Hotline	Government Aid	02-8882-0918 / 1555
6	Safe Spaces PH	Online Abuse & Counseling	0917-544-8624 / safes...
7	Philippine Mental Health Association (PMHA)	Mental Health Counseling	02-8929-6541 / pmha....
8	Kababaihan Para sa Kaunlaran (KPK)	Women's Rights & Support	0917-123-4567
9	National Center for Mental Health (NCMH) Crisis Hotline	Suicide / Mental Health	0917-899-8727 / 02-8...
10	Lingap Center for Women & Children	Counseling / Shelter	0927-555-0198

Figure 13: *Support Resources Page*

Admin Side

- **Admin Dashboard -** Displays all submitted reports and allows administrators to manage and oversee user cases.



ReportID	UserID	FullName	HarassmentType	Description	EvidencePath	Status	DateSubmitted
1	2	Sarina Mae Laurel	Cyber	Sample only		Completed	3/19/2026 11:12...
2	9	Shane Villanueva	Verbal	Example only	C:\Users\Daniel\D...	Ongoing	3/20/2026 12:56...

Figure 14: *Admin Dashboard*

- **Status Update** - Enables administrators to change the status of a selected case (e.g., Pending, Ongoing, Completed).

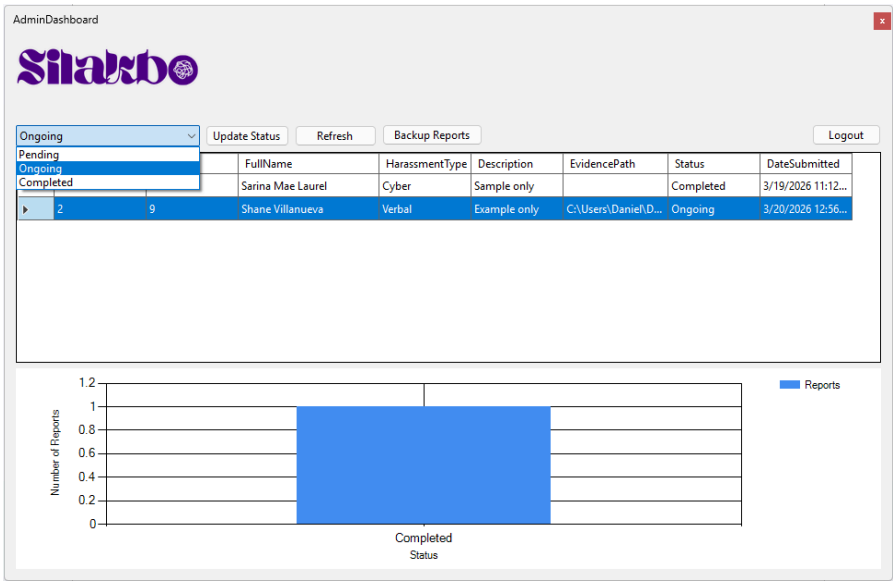


Figure 15: *Status Update*

- **Backup Reports** - Allows administrators to export all report data into a JSON file for backup and record-keeping purposes.

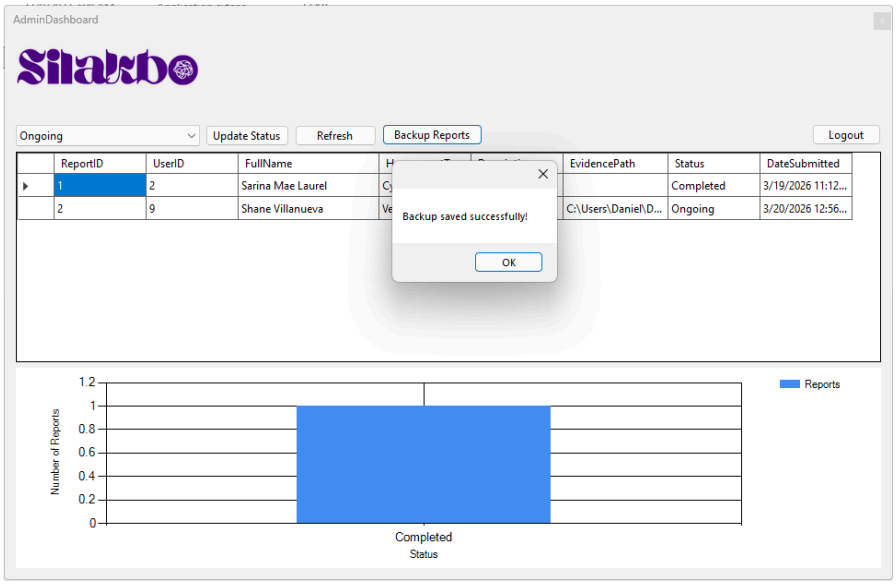


Figure 16: *Backup Reports*

Academic Integrity:

This project is based on the group's original idea and design. AI tools (ChatGPT and Gemini) were used mostly for debugging, complex logic assistance, and improving grammar. All AI suggestions were reviewed and integrated by the group.